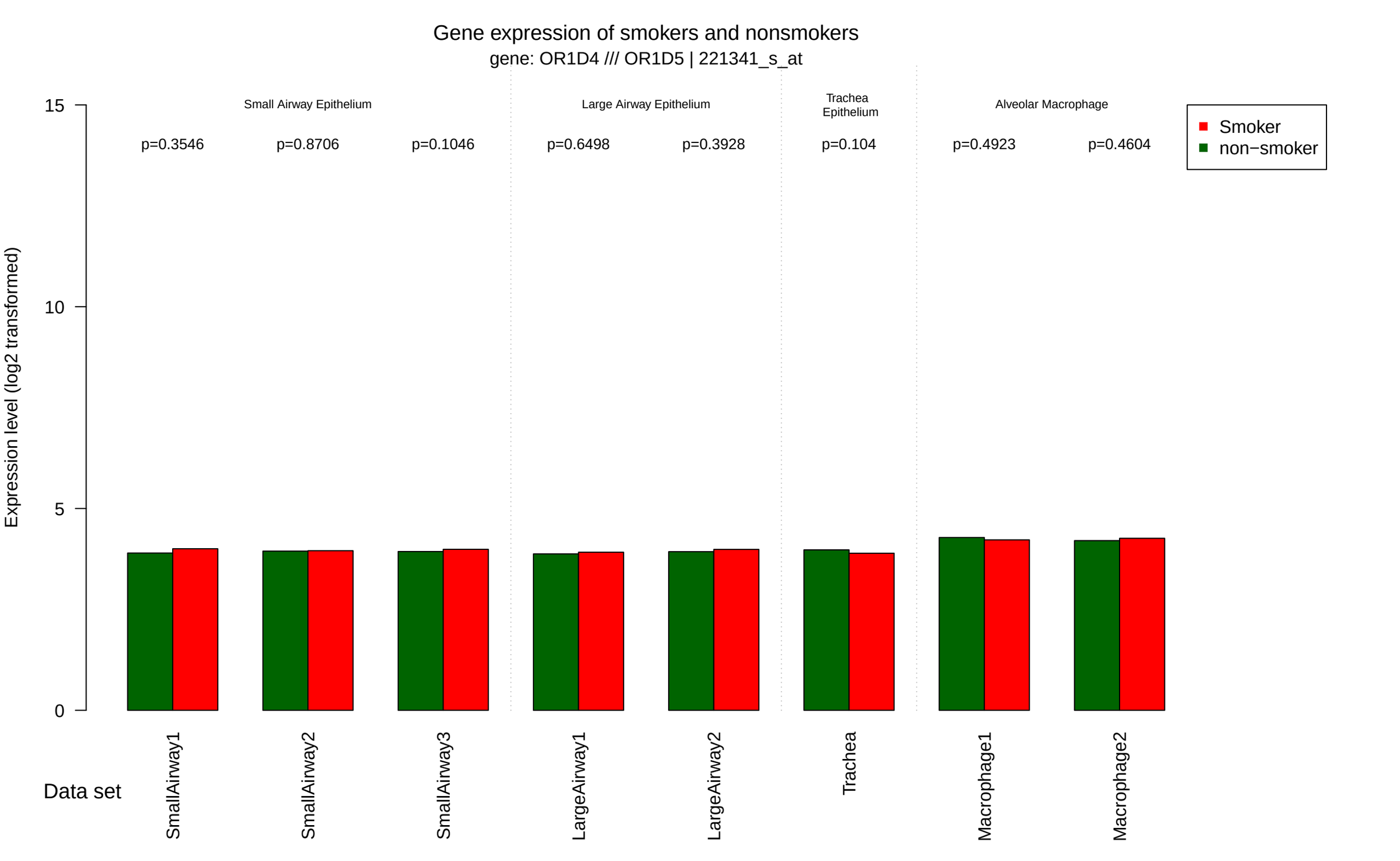
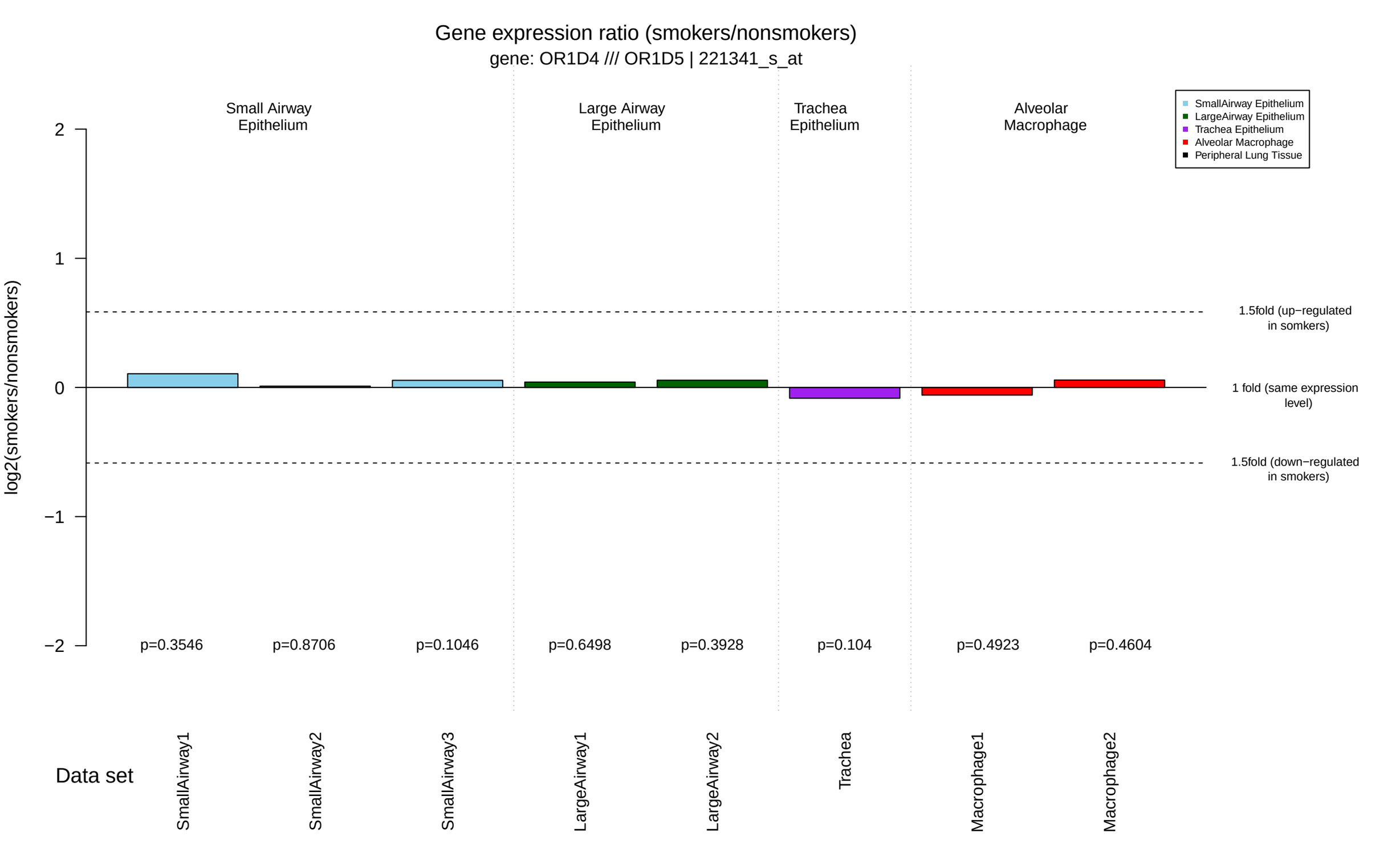
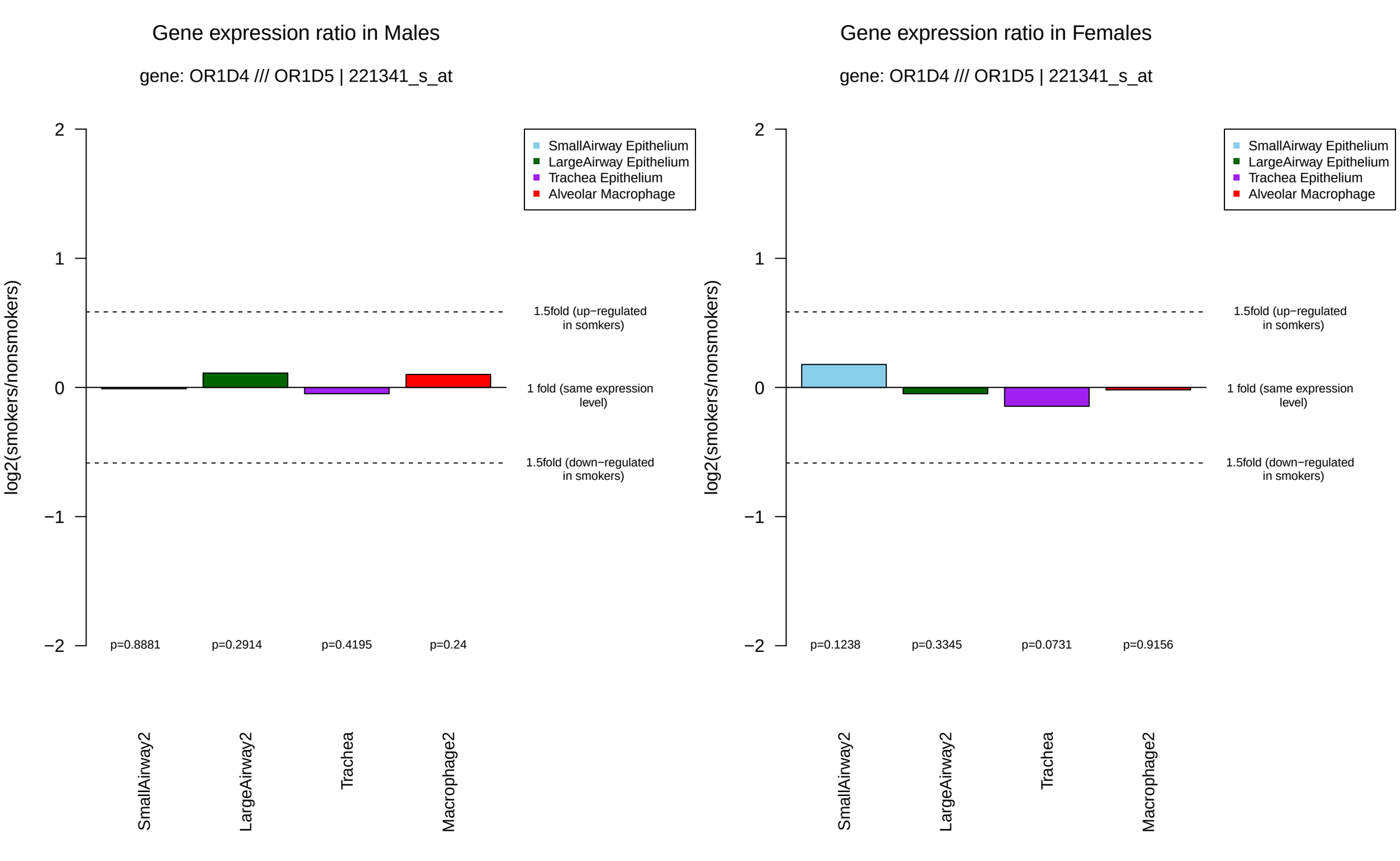
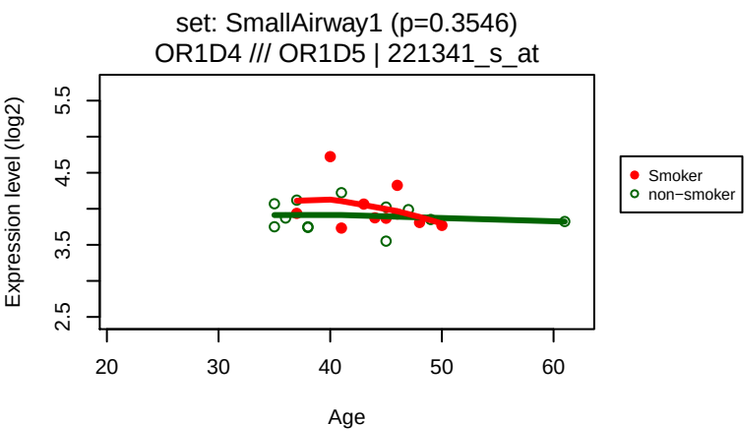
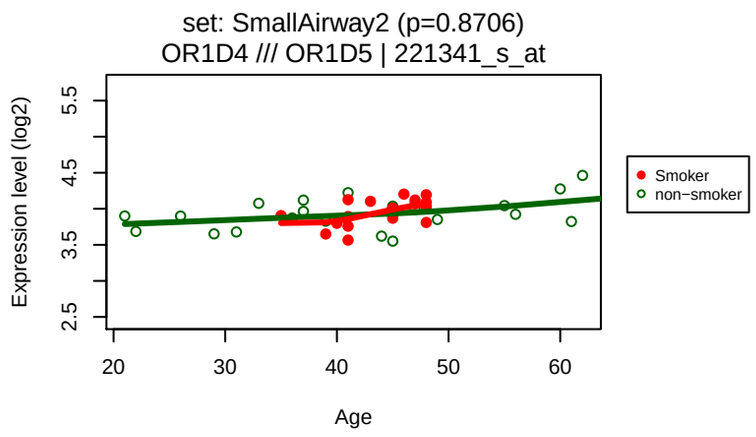
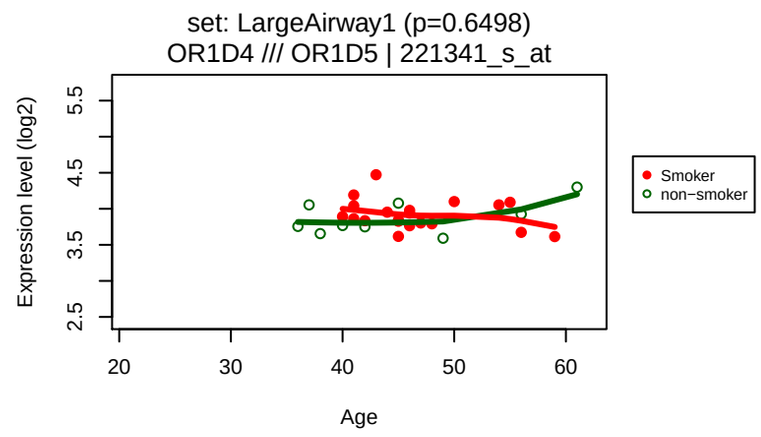

Gene expression of smokers and nonsmokers
gene: OR1D4 /// OR1D5 | 221341_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3546
p=0.8706
p=0.1046
p=0.6498
p=0.3928
p=0.104
p=0.4923
p=0.4604
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

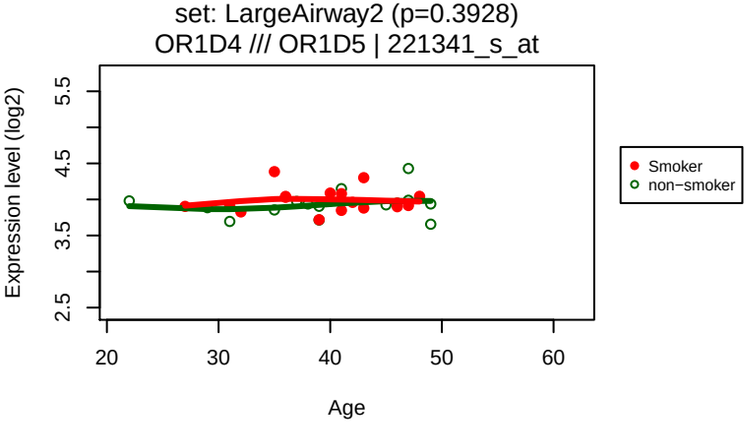

Gene expression ratio (smokers/nonsmokers)
gene: OR1D4 /// OR1D5 | 221341_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3546
p=0.8706
p=0.1046
p=0.6498
p=0.3928
p=0.104
p=0.4923
p=0.4604
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

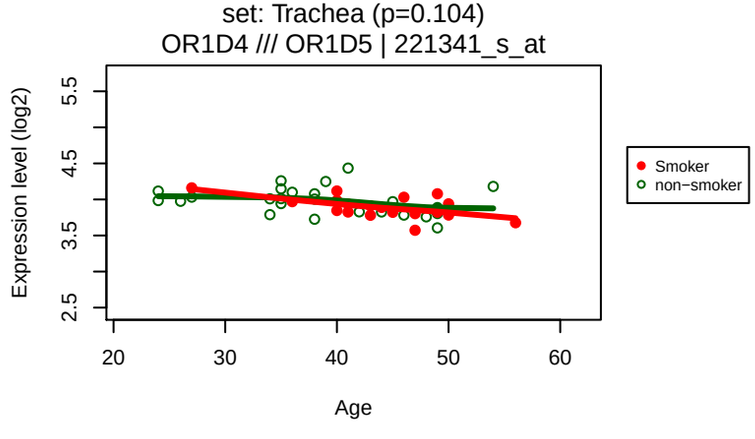

Gene expression ratio in Males
gene: OR1D4 /// OR1D5 | 221341_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8881
p=0.2914
p=0.4195
p=0.24
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR1D4 /// OR1D5 | 221341_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1238
p=0.3345
p=0.0731
p=0.9156
SmallAirway2
LargeAirway2
Trachea
Macrophage2

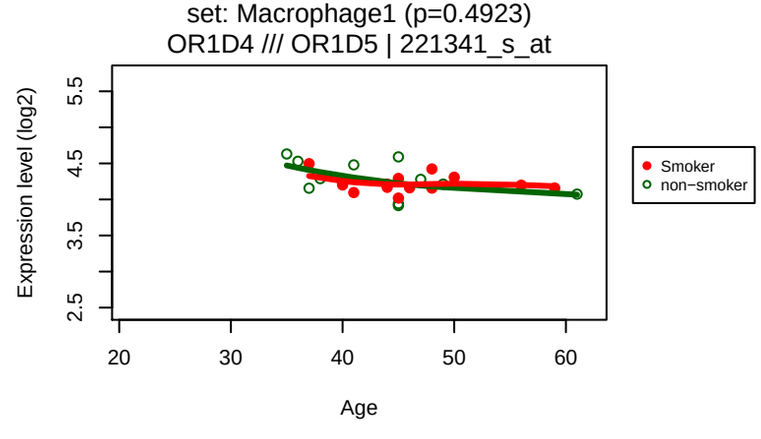

set: SmallAirway1 (p=0.3546)
OR1D4 /// OR1D5 | 221341_s_at
Expression level (log2)
Age
Smoker
non−smoker

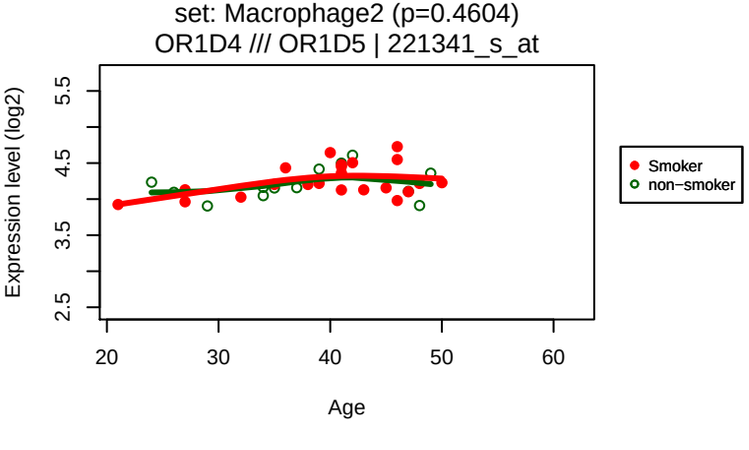

set: SmallAirway2 (p=0.8706)
OR1D4 /// OR1D5 | 221341_s_at
Expression level (log2)
Age
Smoker
non−smoker

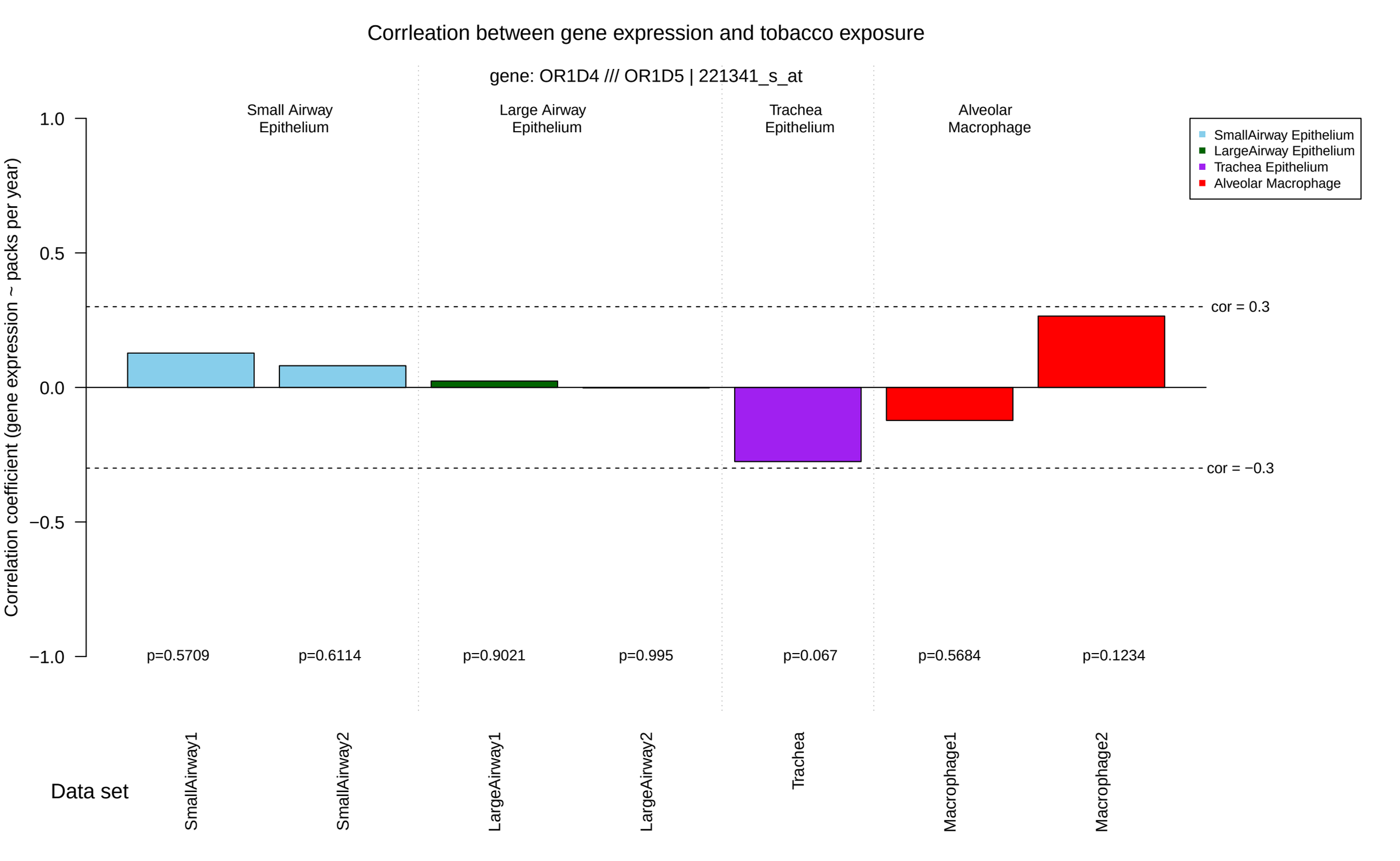

set: LargeAirway1 (p=0.6498)
OR1D4 /// OR1D5 | 221341_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3928)
OR1D4 /// OR1D5 | 221341_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.104)
OR1D4 /// OR1D5 | 221341_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4923)
OR1D4 /// OR1D5 | 221341_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4604)
OR1D4 /// OR1D5 | 221341_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR1D4 /// OR1D5 | 221341_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5709
p=0.6114
p=0.9021
p=0.995
p=0.067
p=0.5684
p=0.1234
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2